

Quasi-Static Magnetic Flux Tube Theory Note - Rev C

1. Flux-tube concept

A flux tube confines magnetic flux. Closed magnetic-field lines are represented by connected lumped magnetic elements. At a magnetic node, signed fluxes satisfy continuity. In quasi-static models, field quantities are complex phasors while a linear reluctance remains real.

2. Fundamental relations

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Phi = integral_A B_n dA
Vm = integral_l H_n dl
Rm = Vm/Phi
For constant A and l: Rm = l/(mu0*mu_r*A)
B = Phi/A
H_iron = B/(mu0*mu_r)
H_air = B/mu0
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3. Leakage split

The released leakage model is parameterized by sigma such that $\Phi_{\text{gap}}/\Phi_{\text{core}} = 1 - \sigma$ and $\Phi_{\text{leak}}/\Phi_{\text{core}} = \sigma$. Because GAP-1 and LEAK-1 are in parallel, they have equal magnetic potential difference.

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R_leak = R_gap*(1-sigma)/sigma
R_parallel = 1/(1/R_gap + 1/R_leak)
Phi_core = N*I / (R_core + R_parallel)
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4. Induced voltage

For a sinusoidal flux phasor at frequency f , induced-voltage magnitude is $2\pi f N |\Phi|$. Orientation determines the voltage sign/phase reference. The exciting and measuring coils are intentionally defined with opposite induced-voltage orientation in the benchmark.

5. Explicit exclusions

Force interaction is excluded. The baseline benchmark does not use a nonlinear B-H curve, hysteresis, eddy-current loss, or frequency-dependent permeability. A nonlinear inductor example may exist elsewhere but it is not authority for this case.